

Running the CalGames Content Desk

A complete setup and operating guide for volunteers

CalGames 2026 • Woodside High School • October 16–18 • Western Region Robotics Forum

Read this first

This guide assumes you have never run this program, never used OBS, and never touched a video switcher. That is the expected starting point. Nothing here requires you to type commands, edit code, or understand how any of it works.

You do not need to read all of it. Find your job below and read that chapter plus the two short ones before it. The whole thing takes about forty minutes end to end; your chapter takes ten.

IF YOU ARE...	READ CHAPTERS
Setting up on Friday	1, 2, 3, 4, 5
Desk manager / producer	1, 2, 6, 7, 11
On-air talent (announcing, analysis, interviews)	1, 2, 8
Field tech	1, 2, 3, 9
Replay operator	1, 2, 7
Trivia host or arcade scorer	1, 2, 10
Anyone who just needs a screen on a TV	1, 2, 5

The one rule. If something looks wrong on the broadcast, the fix is almost never in this program — it is a person pressing the wrong button, and the person who notices should say so out loud. Nobody gets in trouble for calling out a wrong graphic. The show is a team sport.

Chapter 1 · What this thing is

The content desk is one program that runs on one laptop. Everything else — every screen, every control panel, every phone — is just a web page pointed at that laptop.

That is the single most useful idea in this guide, so it is worth saying twice:

There is one program. Everything else is a browser tab.

So there is nothing to install on the pit monitors, nothing to install on the announcer's tablet, and nothing to install on anyone's phone. You type an address, and a screen appears.

What it does

- Draws the **scoreboard and graphics** that go on the stream and on the TVs in the hall.
- Tells the **announcer** who is playing, what their record is, and what they need to do to earn a ranking point — in words, not jargon.
- Cuts **replays** and **match videos**, and uploads them.
- Runs the **crowd trivia** and the **arcade** side tournament between matches.
- Runs the **house music** in the hall.
- Answers **"when does my team play?"** on any parent's phone.

The three jobs

At minimum this show runs with three people. More is better; three is enough.

Desk manager (producer). Sits at the desk laptop. Drives the graphics, switches cameras, keeps the show moving. If there is only one person who knows how the desk works, it is this one.

On-air talent. Announces, analyses, does interviews. Works off a tablet showing the talent view, which tells them everything they need to say.

Field tech. Owns the connections: is the desk talking to the field, is OBS up, is the stream live, is the recording running. Checks the pre-show list in chapter 3 and fixes what it flags.

If you have more people, the next two to add are a **dedicated switcher** (so the desk manager stops cutting cameras) and a **second talent** (so the announcer has somebody to talk to). After that, a replay operator.

Chapter 2 · Starting the desk

Step 1 — Copy one file

Copy `Ca1GamesContentDesk.exe` onto the desk laptop. Put it anywhere — the Desktop is fine.

That is the whole installation. There is no installer, no admin password, and nothing goes into Program Files.

Step 2 — Double-click it

A black window opens and prints what it is doing. It takes about a minute the first time and a few seconds after that.

Windows may show a blue "**Windows protected your PC**" box. This is normal for any program that has not paid for a code-signing certificate. Click **More info**, then **Run anyway**. If you are not comfortable doing that, get the content desk lead — do not click past a security warning on somebody's word alone, including this document's.

Step 3 — Read the address

When it is ready the window prints something like:

```
READY
http://10.0.100.23:8720/
PIN 0864
```

Write that address on the whiteboard. Every screen in the building is going to need it. Write the PIN there too — it is a door code, not a secret; roughly eight people need it.

The numbers will be different at your event. Use what your window prints, not what is printed here.

Step 4 — Leave the window open

Closing the window stops the desk and every screen in the building goes blank. Minimise it, push it to a second monitor, but do not close it.

To stop the desk deliberately, close the window or press **Ctrl-C** in it. Either one shuts down cleanly and releases the address so the next start works.

If it will not start

WHAT YOU SEE	WHAT TO DO
"Port already in use"	An older copy is still running. The launcher offers to stop it — say yes.
It prints a field-scan failure	Fine. The desk runs without the field; see chapter 9.
Nothing happens at all	The file did not finish copying. Copy it again.
A blue "Windows protected your PC" box	More info → Run anyway. See step 2.

Trying it at home first

You do not need a field, a camera, or an event to practise. When the desk finds no field it asks whether you want a **pretend match**; press **D** and every screen plays a full match on a loop with realistic scores.

After the first run, two files appear in the desk folder:

- **START-DESK.cmd** — the real thing.
- **START-PRACTICE.cmd** — the pretend match.

Both are double-click. Practise on the second one as much as you like; it cannot touch a real field or upload anything.

Chapter 3 · The pre-show check (field tech)

Open the desk address in a browser on the desk laptop. You get an index page listing every screen. Open **Desk console** (/s/desk). It asks for the PIN once.

At the top of the desk console is the **doors check**. It answers one question: *is this thing ready?* Everything on it is green, amber, or red.

Run it **at doors**, meaning when the audience is let in — not five minutes before the first match.

CHECK	WHAT RED MEANS	WHAT TO DO
Field	The desk cannot see the scoring system	Chapter 9
OBS	The desk cannot drive the camera switching	Chapter 4
Recording	Nothing is being recorded, so there will be no replays or videos	Check the capture device is plugged in
Disk	Less than two hours of recording space	Free up space or change disks now, not at lunch
Publishing	YouTube is not set up	Videos will not upload. Talk to the content desk lead
Microphones	A mic is muted in OBS	Unmute it
House audio	The music machine is not connected	Chapter 10

Amber is not red. Amber means "this will work, but you should know". A missing Blue Alliance key is amber: videos still upload, they just do not get linked.

Everything red on this list will be noticed by somebody in the audience within an hour. That is the entire point of running it early.

Chapter 4 · Setting up OBS

OBS Studio is the free program that actually sends the video to YouTube. The content desk does not stream anything itself — it draws the graphics, and OBS puts them on top of the camera picture.

If somebody else has already set up OBS at this event, skip to chapter 5.

Step 1 — Install OBS

Download OBS Studio from **obsproject.com** and install it. Take every default.

Step 2 — Make the scenes

A "scene" in OBS is one arrangement of pictures — a camera, a graphic, a title. You switch between scenes to change what the audience sees.

Create six scenes with **exactly these names**. Capitalisation and underscores matter, because the desk switches them by name:

SCENE NAME	WHAT IT SHOWS
CG_INTRO	The opening title, before the day starts
CG_MATCH	The field camera — the main one
CG_SCORE	The field camera, for the final score reveal
CG_REPLAY	The replay playback
CG_DESK	The analysis desk camera
CG_ARCADE	The video game capture, for between matches

To add a scene: in the **Scenes** box at the bottom left, click +, type the name, press Enter.

If you get a name wrong, the desk's camera buttons will do nothing and nothing will explain why. Check the spelling twice.

Step 3 — Add the match overlay

This is the important one. It is the scoreboard.

1. Select the **CG_MATCH** scene.
2. In the **Sources** box, click + → **Browser**.
3. Name it `over1ay` and click OK.
4. Fill in the box exactly like this:

FIELD	VALUE
URL	http://10.0.100.23:8720/s/program — use <i>your</i> address
Width	1920
Height	1080
Use custom frame rate	leave unticked
Shutdown source when not visible	UNTICK THIS
Refresh browser when scene becomes active	UNTICK THIS

Those last two matter. If either is ticked, the overlay reloads every time you cut back to the camera, and the audience watches the scoreboard flash and re-animate mid-match.

1. Click OK. The scoreboard appears over your camera.
2. Drag it so it fills the frame exactly. Right-click it → **Transform** → **Fit to screen** does this in one step.

Step 4 — Copy the overlay to the other scenes

The scoreboard belongs on the score scene and the replay scene too.

Do **not** add a second Browser Source — that runs a second copy of the overlay and doubles the load on the laptop. Instead:

1. Right-click the `Overlay` source → **Copy**.
2. Select `CG_SCORE`, right-click in the Sources box → **Paste (Reference)**.
3. Do the same for `CG_REPLAY`.

Paste (Reference), not **Paste (Duplicate)**. A reference is the same overlay showing in two places. A duplicate is two overlays, and they will drift out of step with each other.

Step 5 — The telestrator (optional)

If your analyst is going to draw on replays:

1. Select `CG_REPLAY`.
2. + → **Browser**, name it `Telestrator`.
3. URL `http://YOUR-ADDRESS/s/tele`, 1920 × 1080, same two boxes unticked.
4. Drag it **above** the Overlay in the Sources list.

Step 6 — The arcade overlay (optional)

Same recipe on the `CG_ARCADE` scene, with URL `/s/arcade`.

Step 7 — Let the desk drive OBS

This lets the desk manager cut cameras from the desk console instead of reaching for OBS.

1. In OBS: **Tools** → **WebSocket Server Settings**.
2. Tick **Enable WebSocket server**.
3. Click **Show Connect Info** and note the **Server Password**.
4. Click OK.

Then tell the desk about it. Ask the content desk lead to start the desk with OBS enabled — this is one of the few things that needs a person who is comfortable with a command line, and it is a two-minute job.

You can run the entire show without this. The desk manager just switches scenes in OBS by hand instead. Everything else works identically.

Step 8 — Check it

Load a practice match (chapter 6) and look at your OBS preview. You should see the scoreboard over the camera, with the clock counting.

If the overlay is **black instead of transparent**, the Browser Source lost its transparency. Delete it and add it again — do not add a Color Correction filter, which is the usual wrong fix.

A note for events with a hardware switcher

If you are going through an ATEM or similar rather than OBS, add `?key=luma` to the overlay URL:

```
http://YOUR-ADDRESS/s/program?key=luma
```

That swaps the transparent background for solid black so a luma key works. Without it the graphics will key out into nothing. On the OBS path, leave it off.

Chapter 5 · The screens around the hall

Every TV, monitor and phone in the building shows a web page. None of them need the PIN.

The easy way

On the monitor, open a browser and go to the desk address. Click **Pick a screen**. Choose from the list. Press **F11** for full screen.

That is it. The page remembers full screen, so if the monitor loses power and comes back, F11 once and it is right again.

The screens, and where each one belongs

SCREEN	ADDRESS	PUT IT
Program with the field feed	/s/watch?screen=program	Pit monitors, concession TVs — the broadcast picture without needing OBS
Side screen	/s/watch?screen=side	The big TVs beside the field. Shows who is on deck and the current rankings, rotating on its own
When do we play?	/s/watch?screen=next	Anywhere parents gather. Also the QR code on the side screens
Trivia	/s/watch?screen=trivia	A monitor the crowd can see between matches
Arcade	/s/watch?screen=arcade	Over the video game capture

Side screens specifically

The side screens are the ones most people in the hall will actually look at. They rotate on their own between "who plays next" and the rankings, with no operator.

They are designed to be read from thirty feet away, which means the type is large and there is not much on screen at once. That is deliberate — do not try to fit more on.

If a side screen goes blank or shows "reconnecting", the desk laptop went away. The screen recovers by itself when the desk comes back; you do not need to touch the monitor.

The field feed behind the program screen

/s/watch?screen=program shows the overlay on top of a live picture of the field. That picture comes from a setting the content desk lead fills in (kiosk.fieldStreamUrl). If it is blank you get the overlay on the CalGames backdrop instead, which looks intentional and is fine.

Chapter 6 · The desk console (desk manager)

Open `/s/desk` . Enter the PIN once.

This is the main control panel and it is built to be driven by keyboard. The buttons all say what they do; this chapter covers the order you use them in.

The rhythm of a match

The field drives most of this by itself. What follows is what you do when the field connection is working — which is the normal case.

WHEN	THE DESK DOES THIS BY ITSELF	YOU DO THIS
Match is queued	Shows the alliance overview with robot photos	Cut to <code>CG_MATCH</code>
Field arms the match	Score bar comes in	Nothing
Match starts	Clock runs, scores update live	Watch the show, not the desk
Buzzer	Clock stops	Cut to a wide, or to the desk
Referees finish	—	If a card was given, chapter 7
Score posts	Final score screen appears	Cut to <code>CG_SCORE</code>
Gap before the next match	—	Sponsor, trivia, arcade, or an interview

If the field connection is down, you press those beats yourself: the match lifecycle buttons across the top of the console are Preview, Arm, Start, End, Post score, Abort. The graphics behave identically.

Keyboard shortcuts

The console is built to be driven by keyboard, because reaching for a mouse during a match is how you miss the thing you were watching for.

The match beats, along the number row:

KEY	DOES
1	Preview — alliance overview
2	Arm — score bar in
3	Start the match clock
4	End — buzzer
5	Post the score
0	Abort

Everything else worth learning on day one:

KEY	DOES
Space	Mark a replay moment. The one to learn first
T	Show the lower third (the name strip)
Y	Hide it
B	Blank the program screen

And, only for when the field connection is down and you are scoring by hand — red on the left of the keyboard, blue below it, the way they sit on the field:

KEY	RED	KEY	BLUE
Q	+1 fuel	A	+1 fuel
W	+5 fuel	S	+5 fuel
E	+10 tower	D	+10 tower

Anything you type this way renders **outlined** on air, so the audience is never shown a guess dressed as a result. See chapter 8.

Camera scenes are buttons on the console rather than keys, because cutting to the wrong camera by fumbling a key is worse than reaching for a mouse.

The console never steals a key while you are typing in a box, and it leaves Ctrl and Alt combinations to the browser — Ctrl+O resets zoom rather than aborting the live match.

Putting people on camera

When somebody sits down for analysis or an interview:

1. In the **On camera** section, tick their name in **Saved profiles**. If they are new, type their name and role and press **Add**.
2. Check the order in **On air, in screen order**. Seat 1 is the audience's **left**. Use the arrows to move people.
3. Press **Put on air**.

Their name cards appear across the bottom of the screen, spaced evenly.

If any of them is a student, tick "Student". Their card then shows a first name and a family initial — "ALEX R." — instead of a full name. Everyone in the hall still knows exactly who they are; a search engine does not, and the video is public forever. Adults doing a job under their own name — mentors, coaches, the commentators — stay in full.

If you put students on with no adult on the panel, the console tells you so. Check the shot before you cut to it: YouTube can disable chat or take a stream down over minors on camera without a visible adult, and getting that wrong ends the stream, not the clip.

Press **Clear** when they get up.

Status cards and safety messages

Two different things, deliberately.

- **Status card** — "Field delay", "Back in ten minutes". Explains a pause. Retires itself.
- **Safety message** — covers every screen in the building, and stays until you clear it. This is for an evacuation or a hold. It is not for a long queue.

Both go to the program, the side screens and every phone at once. When you send a safety message the console goes red so that whoever walks up next can see one is live.

Chapter 7 · Cards, replays, and the awkward moments

When a team gets a card

The field usually tells the desk, and a small mark appears next to that team's number: an outlined square for yellow, a filled block for red. Different *shapes*, not just different colours, so it survives a compressed stream.

For the announcement itself, use the **card call** screen. This puts a full-screen card up after the buzzer and before the score, so the announcer has something on air while they explain it.

1. Pick the team from the list — it only offers the six who just played.
2. Pick yellow or red.
3. Type the reason **in the words the announcer will say**. It goes on screen 200 pixels tall in front of the whole gym.
4. Press **Put the card up**.
5. Press **Clear** before the score reveal.

If the desk says the card is still up after you press Clear, press Clear again. It will tell you if it failed; it will not leave you guessing.

Replays

Press **M** any time something interesting happens. You do not need to be exact — the mark lands a couple of seconds *before* you pressed, because everybody presses late.

The replay console (`/s/replay`) lists the marks. The desk has also marked things on its own: scoring bursts, lead changes, the end of auto.

Surrogates

If a team is playing a surrogate match — one that does not count for their record — the graphics say so. Nothing for you to do; it is there so the audience is not told a team lost a match that was never theirs to lose.

Chapter 8 · The talent view (on-air talent)

Open `/s/talent` on a tablet. It asks for the PIN once.

Everything on this page is something you can say out loud. That is the design rule: no jargon, no raw numbers that need translating.

What is on it

- **Who is playing**, with team names, records, and ranks.
- **What each alliance needs for a ranking point**, in a sentence. Not "ENERGIZED: 87/100" but "thirteen more fuel for Energized".
- **Pronunciation notes**, where somebody has written one in.
- **Cards a team is carrying**, so you are not surprised.
- **Whether a team is a surrogate**, so you do not say the wrong thing about their record.
- **How the day is running** — ahead or behind schedule, in minutes.

A number with an outline around it

If a score renders as an **outline instead of solid**, it means the desk is *guessing* — usually because the field connection dropped and somebody at the desk is typing scores in by hand.

Do not read an outlined number as final. Say "unofficial" or wait. When the real score posts it goes solid.

This is the single most important thing on this page. The graphics will never present a guess as a fact, and neither should you.

Interviews

The desk manager puts your guests' names on screen. Give them a moment to do it before you throw to the interview — the cards animate in, and starting the question before the name lands means the audience never learns who is talking.

Chapter 9 · The field connection (field tech)

The desk can read the scoring system so the graphics follow the real match without anyone pressing anything.

It is read-only, by construction

The desk **cannot** start a match, abort a match, change a score, or touch the field in any way. That is not a policy — the connection is built so that those requests do not exist. You can hand this document to the FTA and the head referee and that sentence is the whole of it.

Setting it up

Usually nothing. The launcher looks for the field on the standard event network when it starts, finds it, and says so.

If it does not:

- 1. Check the desk laptop is on the **same network** as the scoring system. Event Wi-Fi and the field network are often not the same thing.
- 2. Check the scoring computer is on and its display is up.
- 3. Restart the desk (close the window, double-click again). The scan runs at start.

If it stays down

The show runs anyway. Tell the desk manager, who switches to driving the match beats by hand (chapter 6). Everything looks identical to the audience except that scores are typed rather than read — and those render outlined, so nobody is misled.

This is worth rehearsing once on Friday. A field connection that drops during finals is a much better experience for everyone if the desk manager has done it before.

The other connections you own

THING	HOW YOU CHECK IT	IF IT IS DOWN
OBS	The doors check, and the OBS window itself	Chapter 4
Stream	YouTube Studio shows it live	Restart the stream in OBS; the desk is unaffected
Recording	Doors check says Recording	Replays and videos stop; the live show is unaffected
Disk space	Doors check says Disk, in <i>hours</i>	Free space now

Chapter 10 · Trivia, the arcade, and the music

Crowd trivia

Open `/s/triviadesk`. The job is three buttons: **Open**, **Reveal**, **Next**.

The crowd joins on their phones from the QR code the trivia screen shows. There are more than thirty questions loaded already, in rounds, so you can play a round after one match and pick up a different round after the next. Scores carry across the day.

A round that has been played through will refuse to replay — that is on purpose, because replaying one pays everybody twice and quietly corrupts the leaderboard.

Names the crowd types are screened before they go on screen. Some will still get through that a reasonable person would not want on a projector; you can remove any name from the host console.

The arcade

Open `/s/arcadedesk`. Score the game sets by hand: two players head-to-head, or three-to-four free-for-all. The overlay shows the standings.

Each info box on the arcade overlay can be shown or hidden on its own, so you can clear the screen when the game itself is the thing to watch.

House music

This is the one part that runs on a different computer.

The music machine is whatever laptop is plugged into the hall's PA. Open `/s/house` **on that machine** and leave it open.

Why a separate machine: music played in the hall must not reach the stream. A recorded track on a YouTube video can get the whole archive muted or blocked, and the archive is what teams watch afterwards. Keeping the music on a machine that is not wired to the stream makes that structural rather than a thing somebody has to remember at 4pm on a Sunday.

The page refuses to run inside OBS for the same reason. If you see it refuse, that is it working.

From the desk console the manager can pause the music, resume it, duck it under an announcement, fire a team's walk-up song, or hand the room to the video game audio — all without leaving their seat.

Chapter 11 · When something goes wrong

Ordered by how often it happens.

WHAT YOU SEE	ALMOST CERTAINLY	FIX
A screen says "reconnecting"	The desk laptop went to sleep or the window got closed	Wake it / start it again. Screens recover on their own
The overlay is frozen	The Browser Source has "shutdown when not visible" ticked	Chapter 4, step 3
Scores are not updating	Field connection dropped	Chapter 9. Drive it by hand meanwhile
A score has an outline around it	The desk is guessing, and saying so	Nothing. It goes solid when the real score posts
Camera buttons do nothing	OBS is not connected, or a scene is misspelled	Chapter 4, steps 2 and 7
Music is coming out of the stream	The house page is on the wrong machine	Chapter 10 — this one matters
No robot photos	None uploaded for those teams	Fine. It falls back to a gold team number
A wrong name on screen	Somebody typed it	Fix it in the profile list and press Put on air again
"Port already in use" on start	An old copy is still running	The launcher offers to stop it. Say yes

The universal fix

Close the black window. Double-click the desk again. It comes back in a few seconds, remembers the day — cards, coverage, the run of show, sponsor counts — and every screen in the building reconnects on its own.

You do not lose the day by restarting. That is deliberate, and it was built because laptops get their power cables kicked out.

When to escalate

Get the content desk lead for: anything involving credentials, uploads going to the wrong channel, or a message on screen that should not be there and will not clear. Everything else on this page is yours to fix.

Appendix A · Every screen, in one table

Replace ADDRESS with what the desk window printed.

On air — these go into OBS

SCREEN	URL	NOTES
Program overlay	ADDRESS/s/program	1920×1080 Browser Source. The scoreboard
Telestrator render	ADDRESS/s/tele	Transparent. Layer over the replay
Arcade overlay	ADDRESS/s/arcade	Over the game capture
Trivia overlay	ADDRESS/s/trivia	Crowd game
Side screen	ADDRESS/s/side	For venue TVs, not for OBS

Run the show — these need the PIN

SCREEN	URL	WHO USES IT
Desk console	ADDRESS/s/desk	Desk manager
Talent view	ADDRESS/s/talent	On-air talent
Replay console	ADDRESS/s/replay	Replay operator
Telestrator pad	ADDRESS/s/draw	Analyst's tablet
Arcade console	ADDRESS/s/arcadedesk	Arcade scorer
Trivia host	ADDRESS/s/triviadesk	Trivia host
House audio player	ADDRESS/s/house	On the music machine only
Head referee review	ADDRESS/s/var	Head referee. Read-only
Phone remote	ADDRESS/s/remote	Anyone away from the desk

Before the event

SCREEN	URL	WHO USES IT
Team media	ADDRESS/s/media	Whoever has the robot photos
Post-match cards	ADDRESS/s/cards	Social media

For everyone — no PIN

SCREEN	URL	NOTES
Pick a screen	ADDRESS/s/watch	Hand this to anyone setting up a monitor
When do we play?	ADDRESS/s/next	Parents, on their own phones
Trivia play	ADDRESS/s/quiz	The crowd's phones

Appendix B · The Friday setup list

Print this. Tick it off.

AV rack

- ☐ Desk laptop on the event network, plugged into power
- ☐ CalGamesContentDesk.exe copied across and run once
- ☐ Address and PIN written on the whiteboard
- ☐ Field connection confirmed (chapter 9)

OBS

- ☐ OBS installed
- ☐ Six scenes created, names checked twice (chapter 4, step 2)
- ☐ Program overlay added to CG_MATCH, both boxes unticked
- ☐ Overlay pasted as *reference* into CG_SCORE and CG_REPLAY
- ☐ Stream key entered, test stream run and stopped
- ☐ Recording tested — record thirty seconds and play it back

Screens

- ☐ Side screens up, full screen, rotating
- ☐ Pit monitors up
- ☐ Trivia monitor up, QR readable from where the crowd stands

Audio

- ☐ Music machine identified and /s/house open on it
- ☐ Announcer mics tested through the stream, listening on a phone
- ☐ Confirmed music does **not** appear on the stream

People

- ☐ Everyone has read their chapter
- ☐ Desk manager has run a practice match end to end
- ☐ Desk manager has run one match with the field disconnected, on purpose
- ☐ Talent has seen the talent view and knows what an outlined number means

Content

- ☐ Robot photos uploaded (/s/media)
- ☐ Sponsors entered
- ☐ Run of show entered

☐ Walk-up songs in place for playoff introductions

Appendix C · Words this guide uses

WORD	MEANS
Browser Source	An OBS source that shows a web page. How the graphics get on the stream
Scene	One arrangement of pictures in OBS. You switch between them
Overlay	The graphics on top of the camera picture: score bar, clock, names
Lower third	The name strip along the bottom
Program	What the audience is seeing right now
Cut	Switch instantly from one scene to another
On deck	The match after the one being played
Surrogate	A team playing an extra match that does not count for their record
Luma key	A way hardware switchers lay graphics over video, using black as transparent
Doors	When the audience is let in. The deadline the pre-show check exists for

*This guide is part of the CalGames Content Desk, which is open source under the Apache 2.0 licence.
Corrections are welcome and genuinely useful: if something here was wrong or confusing while you were
standing in a gym trying to make it work, that is a bug in this document.*